Split-half correlations on earlier or shorter windows

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1 Takeaways

- Split-half correlations for earlier and shorter windows align with the original split-half correlations.
- IT is an exception, since earlier and shorter windows produce very low split-half correlations.
- The between-family trend stays similar, though IT distances are lower relative to V4 (likely due to the difficulty in matching correlations). The trend also appears in noisy textures.
- The between-sample trend changes, with V1 trending higher, possibly higher than V4 and IT. V2 remains the "best".
- The changes require more investigation, as the correlations are harder to match with the new averages. The results may be confounded with the distribution of the reliabilities.

2 Split-half correlations with other onset time schemes

Here, I check if the previous between-family and between-sample results are consistent if the onsets are shifted earlier or if the averaging window is shorter.

The calculations for the new split-half correlations can be found in `scripts/split_corrs_earlier_onsets.R` and associated bash script.

2.1 Comparison of original onsets and earlier onsets over shorter windows

To capture initial firing, I calculated split-half correlations over earlier onset times with shorter windows (100 ms vs. the original 150 ms). These earlier, shorter split-half correlations are fairly well correlated with the original split-half correlations (Figure 1). However, in IT, the earlier, shorter correlations tend to be closer to zero, which is not surprising considering 60 to 160 ms barely captures the averaged response curve (not shown).

The distributions of the correlations for each condition can be seen in Figure 2. For V1, taking earlier and shorter windows appears to slightly increase the split-half correlations. This trend reverses for the other regions.

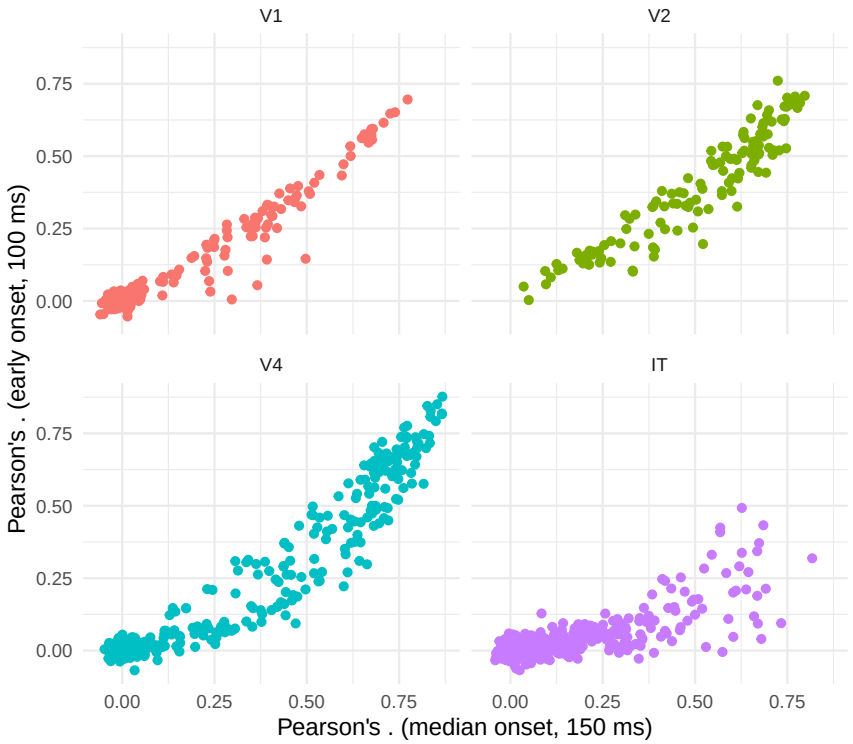


Figure 1: Scatter plots comparing the per-site split-half correlations when onset times are earlier with 100 ms window vs. median with 150 ms window.

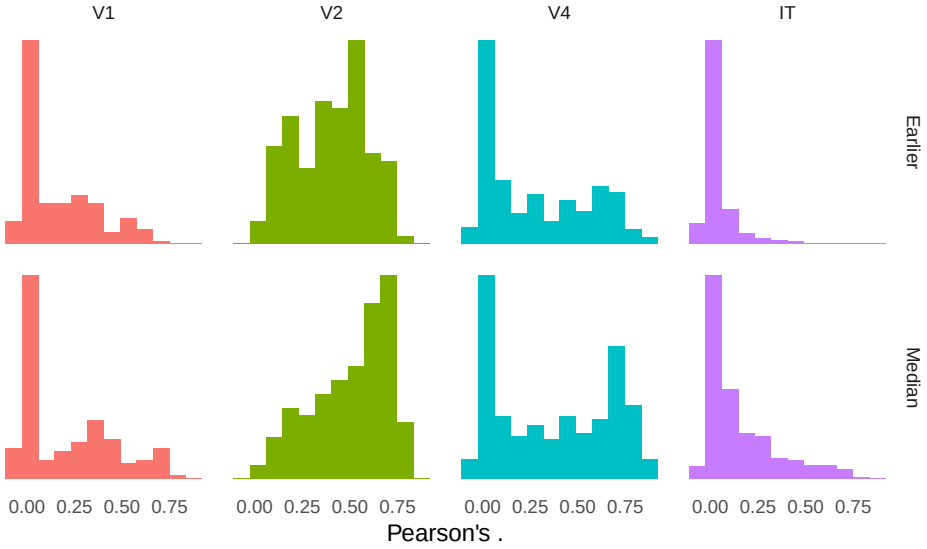


Figure 2: Histograms comparing the per-site split-half correlations when onset times are earlier with 100 ms window vs. median with 150 ms window.

3 Between-family distances

The IT reliabilities are pretty low, so it's not easy to get matched-reliability results with the previous method (selecting sites with correlations in some interval). In Figure 3, IT dips below V4, likely as a result of the low reliabilities. Otherwise, V1 < V2 < V4, as expected.

Unlike the previous results, the trend also emerges in noisy images.

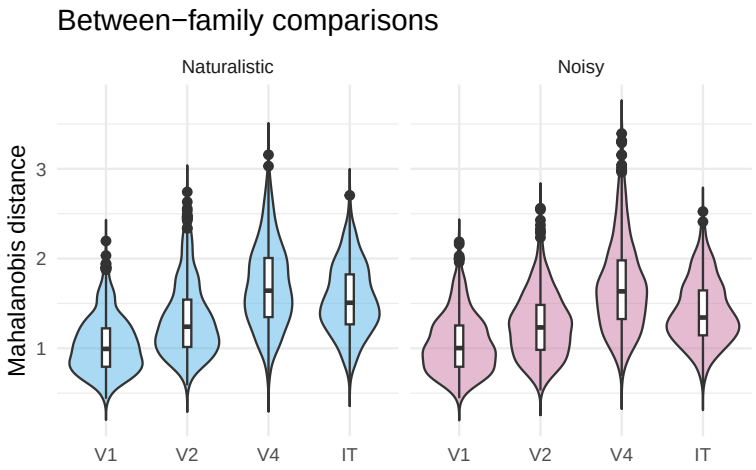


Figure 3: Mahalanobis distances in pairwise family comparisons with earlier, shorter windows.

4 Between-sample distances

Unlike the previous results, the V1 results seem to be better than V4 and IT. V2 remains the "best" (Figure 4).

5 Between-sample comparisons

6 References

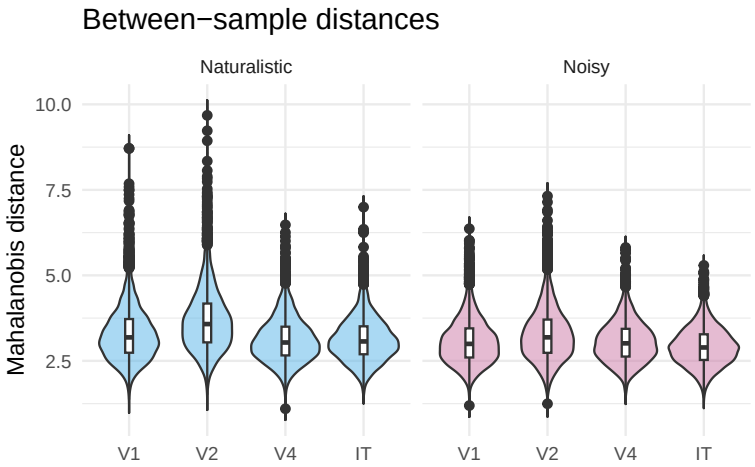


Figure 4: Mahalanobis distances in pairwise sample comparisons with earlier, shorter windows.